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Master in Business Analytics & Data Science

MASTER'S THESIS

Mean-Variance Dynamics and Market Efficiency in Silver

A Multi-Model Forecasting Framework

Asier Ugarteche Perez

Supervisor: Prof. Dae-Jin Lee

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Abstract

This thesis asks how much of silver’s short-term price behaviour can be forecast, and frames the question through the efficient-market hypothesis (EMH). Silver is modelled at the weekly (Friday-to-Friday) horizon over 2015–2026, and the analysis is factorised into the two moments: the conditional *mean* of returns and the conditional *variance*—efficiency constrains the former but not the latter. For the mean, a deliberate ladder of models of increasing flexibility—ARIMA/ARIMAX, VAR, mixed-data-sampling (MIDAS) regressions, random forests, gradient boosting, and an LSTM—is tested against a common random-walk-with-drift benchmark, expanding the information set from silver’s own past (weak-form EMH) to a broad set of public information (semi-strong-form EMH): cross-asset returns, silver futures positioning, macroeconomic releases, technical indicators, and sentiment from Reddit and financial news. For the variance, weekly realised volatility is modelled with GARCH and the heterogeneous autoregressive (HAR) model, and the question of interest is whether sentiment improves the forecast. The central finding is a single coherent statement: no model beats the drift benchmark for the conditional mean by a significant margin, so weak- and semi-strong-form efficiency cannot be rejected—yet the conditional variance is strongly and significantly forecastable, with sentiment adding a small but robust increment. Predictable volatility alongside an unpredictable mean is exactly what the EMH permits, so the two results are complementary rather than contradictory.

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1 Introduction

Silver lives a double life. It is an industrial input—roughly half of annual demand goes into photovoltaics, electronics, brazing alloys, and medical uses, tying its price to the global manufacturing cycle—and, at the same time, a monetary metal held as a store of value, an inflation hedge, and a more volatile cousin of gold. On top of both sits a retail following that can sometimes turn silver into a meme trade: in late January 2021, Reddit’s `r/WallStreetSilver` drove spot from roughly \$25 to \$30 an ounce within days before it faded. This blend of industrial demand, monetary hedging, and episodic retail attention makes silver an unusually rich laboratory for the oldest question in empirical finance: how much of a price can be forecast?

That question has a canonical answer, and it is mostly negative. The efficient-market hypothesis (Fama, 1970, 1991) holds that prices already embed available information, so successive price changes are approximately unpredictable. It is conventionally stated in three nested forms, graded by the information set a forecaster is allowed to use:

- **Weak form** — past prices and returns carry no exploitable signal.
- **Semi-strong form** — no *public* information does either: not prices, fundamentals, news, or macroeconomic data.
- **Strong form** — not even *private*, inside information does.

Each form nests the one before it, so the claims grow progressively stronger. For a liquid, globally traded metal, weak- and semi-strong-form efficiency are the natural hypotheses to take to the data, and they are the two this thesis tests.

Empirically, silver’s *price* behaves like a random walk with drift: it is non-stationary, and shocks accumulate rather than mean-revert (Figure 1). Returns, however, are stationary—they fluctuate around a stable long-run mean rather than wandering without bound. The object worth modelling is therefore not the price level but its increment, the weekly log-return $r_t = \log P_t - \log P_{t-1}$; stationarity is what makes the forecasting problem well-posed, giving a fixed quantity to predict and a natural yardstick—that same prevailing average—to judge any forecast against.

The weekly (Friday-to-Friday) horizon is a deliberate choice of frequency, made where the information aligns. Futures-market positioning is reported weekly and macroeconomic releases monthly, while daily sentiment is too sparse and noisy to inform a forecast and daily returns are dominated by microstructure noise; a weekly horizon is coarse enough to average out that noise yet fine enough to leave roughly 600 observations over 2015–2026—enough to train and properly out-of-sample-test the machine-learning models. Weekly realised volatility, the target of the second-moment analysis, is in turn

aggregated from daily returns.

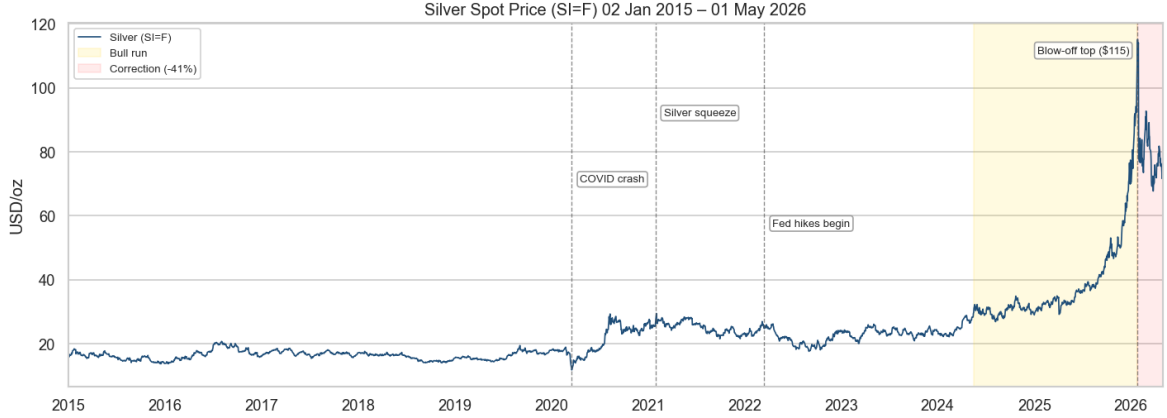


Figure 1: Silver front-month futures (SI=F), daily, January 2015–May 2026. The annotated episodes punctuate the sample: the COVID-19 crash (March 2020), the r/WallStreetSilver squeeze (January 2021), the start of the Federal Reserve’s hiking cycle (March 2022), and the 2024–2026 bull run that carried spot to an all-time high near \$115/oz before a sharp (−41%) correction. The level is non-stationary and shocks accumulate; the thesis models its stationary weekly log-return rather than the level itself.

The efficiency question then becomes whether r_t can be forecast—and silver’s own past offers little to go on. Weekly returns carry almost no linear autocorrelation, so a linear own-history model such as ARIMA has little to grip; a formal model search duly collapses it to a constant mean. Yet the returns are not independent noise either, because their *magnitude* is persistent—volatile weeks cluster together—so structure remains, only in the variance rather than the mean. The thesis follows this thread: it asks first whether silver’s own history forecasts the return, then whether the broad public-information set a real forecaster would possess does—cross-asset returns (gold, the US dollar, copper, equities, oil, and the VIX), futures-market positioning, macroeconomic releases, technical indicators, and retail and news sentiment—and, separately, whether the predictable variance can be modelled.

Research questions. The thesis is built around four questions:

- **Q1 (weak form).** Are weekly silver returns forecastable from their own past?
- **Q2 (semi-strong form).** Does expanding the information set to public covariates help—and does it matter whether the relationship is assumed linear or allowed to be nonlinear?
- **Q3 (the second moment).** Is weekly realised volatility forecastable, and if so by how much?
- **Q4 (sentiment).** Does retail and news sentiment add incremental predictive

content to either the conditional mean or the conditional variance?

Approach. To answer these questions, the analysis estimates a deliberate ladder of models of increasing flexibility against a common benchmark and a unified evaluation protocol. The benchmark throughout is the prevailing-mean “drift” forecast—the random-walk-with-drift, equivalent to ARIMA(0,0,0)—which is the correct benchmark for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008). For the conditional mean, the ladder runs from classical time-series models (ARIMA/ARIMAX, VAR, and MIDAS regressions) through tree ensembles (random forests and gradient boosting) to a recurrent neural network (LSTM); for the conditional variance it runs GARCH and the heterogeneous autoregressive (HAR) model of Corsi (2009) alongside the same tree ensembles.

2 Conceptual Framework

This section fixes the notation used throughout and states, precisely, the null hypothesis that each model class is built to test. The aim is to keep the language exact without overclaiming: the empirical results are framed as tests of the conditional mean and the conditional variance of weekly silver returns, not as claims about the full return distribution.

2.1 Prices, returns, and the random walk

Let P_t be the silver price and $p_t = \log P_t$ the log price. The weekly log-return is the increment in the log price,

$$r_{t+1} = p_{t+1} - p_t, \quad \text{equivalently} \quad p_{t+1} = p_t + r_{t+1}, \quad (1)$$

so the price level and the return are two views of the same process: the level accumulates shocks, while returns *are* the shocks. A random walk with drift writes the level as $p_{t+1} = p_t + \mu + \varepsilon_{t+1}$ with $\mathbb{E}[\varepsilon_{t+1}] = 0$. Silver log prices are best described as *random-walk-like*—non-stationary, with shocks that accumulate—which is the reason the modelling target throughout is the stationary increment r_t rather than the level. The safer wording avoids the *strict* random walk: the innovations are not independent and identically distributed, because their variance clusters (Section 2.4).

2.2 The conditional mean and the information set

Let F_t denote the information available at time t . For the weak-form test, this is silver’s own past, $F_t = \{r_t, r_{t-1}, r_{t-2}, \dots\}$. Returns satisfy the *martingale difference* property

with respect to F_t if their conditional mean is constant,

$$\mathbb{E}[r_{t+1} \mid F_t] = \mu, \quad (2)$$

or, equivalently, the demeaned return $r_{t+1} - \mu$ is a *martingale difference sequence* (MDS),

$$\mathbb{E}[r_{t+1} - \mu \mid F_t] = 0. \quad (3)$$

Past returns do not move the expected next-week return away from its long-run average. This is a statement about the first moment *only*; it does not assert that returns are i.i.d. The semi-strong test expands the information set to public covariates,

$$G_t = \{F_t, \text{ gold, USD, copper, VIX, oil, COT, macro, technicals, sentiment}\}, \quad (4)$$

and asks whether the martingale-difference property survives the larger conditioning set,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu. \quad (5)$$

ARIMA tests whether silver returns are a martingale difference with respect to their own past, Equation (2); ARIMAX and the other models test whether that property survives once the information set is expanded to public covariates, Equation (5). Operationally, these nulls are tested as “does the model beat the prevailing-mean (drift) benchmark out of sample?”—a forecast that beats the drift would contradict the corresponding null.

2.3 Linear and nonlinear alternatives

The alternative to Equation (5) can be specified narrowly or broadly. A linear model entertains an additive alternative,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu + \beta' G_t, \quad (6)$$

so ARIMAX, VAR, and MIDAS ask whether public covariates enter the conditional mean linearly and additively. Nonlinear models entertain the broader alternative

$$\mathbb{E}[r_{t+1} \mid G_t] = f(G_t), \quad (7)$$

in which the conditional mean can respond to thresholds and interactions among the covariates rather than to each one additively—a signal might be invisible in any single predictor yet surface in a combination of them. Random forests, gradient boosting, and the LSTM search this larger space. The logic of the ladder is therefore cumulative: if even flexible nonlinear models fail to beat the drift benchmark out of sample, the

martingale-difference interpretation is *strengthened* rather than merely assumed from linear tests.

2.4 The conditional variance

Efficiency constrains Equation (5); it places no restriction on the conditional variance $\text{Var}(r_{t+1} | F_t)$. The two can diverge: the mean may be (approximately) constant while the variance moves through time, which is exactly what volatility clustering means—a large move today says little about the sign or size of next week’s return but does signal that next week is likely to be more volatile. Strict white noise would require independence in r_t , r_t^2 , and $|r_t|$ alike; the MDS property in Equation (2) restricts only the first of these. Weekly silver returns are close to white noise in the mean but not strict white noise, because squared returns are strongly autocorrelated. The predictable structure is therefore concentrated in the second moment, and the squared return is its natural proxy. By definition it splits into the variance and the squared mean, and at the weekly horizon the latter is negligible:

$$\mathbb{E}[r_t^2] = \underbrace{\text{Var}(r_t)}_{\approx 1.9 \times 10^{-3}} + \underbrace{\left(\mathbb{E}[r_t]\right)^2}_{\approx 7 \times 10^{-6}} \approx \text{Var}(r_t). \quad (8)$$

The squared mean is only 0.4% of the variance, so a squared return is—to that approximation—a model-free measure of variance. This is what the volatility chapter (Section 7) builds on, summing daily squared returns into weekly realised volatility. Figure 2 shows both moments at once for the weekly silver return.

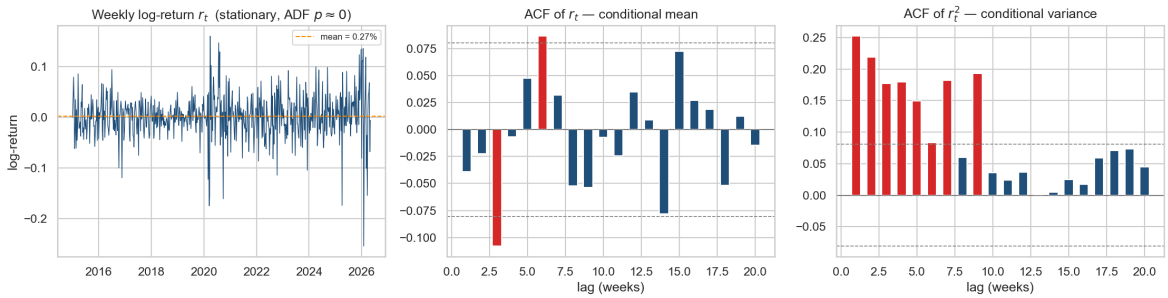


Figure 2: The two moments of the weekly silver log-return, 2015–2026. *Left*: the return series—stationary, fluctuating around a small positive drift (mean $\approx 0.27\%$ per week). *Centre*: the sample autocorrelation of returns lies almost entirely within the 95% white-noise band (dashed), with only isolated lags breaching it—consistent with the martingale-difference null of Equation (2), so a linear own-history model has little to exploit. *Right*: the autocorrelation of *squared* returns is large and positive across the first several lags—the volatility clustering of Section 2.4. The mean is close to unpredictable; the variance is not.

2.5 Direction versus magnitude

A final distinction governs how forecasts are scored. Directional accuracy asks whether $\text{sign}(\hat{r}_{t+1}) = \text{sign}(r_{t+1})$ more often than chance, which is *not* the same as beating the drift benchmark under squared-error loss: a model can call the sign correctly yet fail to improve out-of-sample R^2 . Because efficiency is a conditional-mean claim, the magnitude axis (RMSE, out-of-sample R^2 , and Diebold–Mariano against the drift) is load-bearing, while directional accuracy—and its significance test, Pesaran–Timmermann (Pesaran & Timmermann, 1992)—is a secondary, economically motivated lens. Directional gains are interesting but do not by themselves reject the martingale-difference null.

3 Literature Review

This thesis sits at the intersection of market efficiency, commodity forecasting, volatility modelling, and text-based investor attention. Its central distinction is between two kinds of predictability. The first is *conditional-mean* predictability: whether next week’s silver return can be forecast from past prices or public information. This is the economically load-bearing question for the efficient-market hypothesis. The second is *conditional-variance* predictability: whether the size of next week’s move can be forecast, even if its sign and expected value cannot. The literature suggests that these two forms of predictability should not be treated as contradictory. A market can be efficient in the mean while still displaying persistent volatility, because the EMH restricts expected returns, not time-varying risk.

3.1 Return predictability, market efficiency, and public information

The natural starting point is the martingale logic of Samuelson (1965) and the efficient-market taxonomy of Fama (1970, 1991). In an informationally efficient market, prices incorporate available information quickly enough that price changes are difficult to forecast from that same information. The weak form asks whether past prices and returns contain exploitable signal; the semi-strong form asks whether a broader public information set does. That the weak-form null can be rejected is itself well established—Lo and MacKinlay (1988), with a variance-ratio test on *weekly* returns, reject the random walk for US equity indices—so own-history predictability at the weekly horizon is possible in principle, and testing for it in silver is a genuine question rather than a foregone conclusion. For this thesis, the implication is direct: an ARIMA model tests whether silver’s own return history moves the conditional mean, while ARIMAX, VAR, MIDAS, machine-learning models, and the LSTM test whether public covariates do. If none improves materially on the prevailing-mean drift benchmark out of sample,

the result is evidence in favour of weak- and semi-strong-form efficiency at the weekly horizon.

Commodity markets complicate this simple null, because prices reflect both financial and physical forces. Deaton and Laroque (1992) document the extreme persistence and spike-like behaviour of commodity prices, which makes price levels poor forecasting targets and motivates modelling returns instead. At the same time, commodity futures returns have been linked to hedging pressure, inventories, basis, and risk premia (Bessembinder, 1992; G. Gorton & Rouwenhorst, 2006; G. B. Gorton et al., 2013; Szymanowska et al., 2014). These papers justify including futures positioning and cross-asset information, but they do not imply that a single weekly silver return should be easy to forecast out of sample. The distinction matters: variables may explain risk premia or cross-sectional commodity returns over longer horizons without generating a reliable one-week forecast for silver’s conditional mean.

Silver is also not simply “gold with more volatility.” Precious-metals research shows that gold, silver, platinum, and palladium respond differently to macroeconomic conditions. Batten et al. (2010), for example, find that the macroeconomic drivers of volatility differ across precious metals, while Zhang and Wei (2010) document important linkages between gold and oil. This supports the thesis’s broad public-information set—gold, the dollar, copper, oil, equities, the VIX, rates, inflation expectations, industrial production, and COT positioning—while also warning against assuming that silver’s behaviour can be inferred from gold alone. The CFTC’s Commitments of Traders data are especially relevant because they separate commercial and non-commercial positioning, making them a natural proxy for hedging pressure and speculative sentiment.

The benchmarking literature is equally important. Welch and Goyal (2008) show that many variables that appear to predict returns in sample fail to beat the historical average out of sample. Campbell and Thompson (2008) argue that small gains are possible when forecasts are disciplined by theory, but they retain the historical mean as the central benchmark and out-of-sample R^2 as the natural effect-size measure. This thesis follows that discipline. The drift forecast is not a weak straw man; it is the operational version of the EMH null for a return target.

The adaptive-markets hypothesis of Lo (2004) provides a useful interpretation for unstable or episodic findings. Under the AMH, market efficiency is not a fixed state but an evolutionary outcome shaped by competition, learning, institutional change, and changing trader populations. A temporary directional signal around a retail-attention episode, a crisis, or a commodity squeeze would therefore not be surprising. What matters is whether it survives out of sample, across regimes, and against a benchmark. This is why the thesis combines expanding and rolling windows, subperiod robustness

checks, and a separate interpretation of directional accuracy.

3.2 Mixed-frequency information and nonlinear forecasting models

The semi-strong test raises a practical problem: the information set is observed at different frequencies. Silver and cross-asset prices are daily, COT positioning is weekly, macroeconomic variables are monthly and released with delays, and sentiment can be daily but sparse. MIDAS regression was developed precisely for this kind of problem. Ghysels et al. (2004, 2007) show how high-frequency lags can enter a low-frequency forecast through a parsimonious lag-weight function rather than a large unrestricted lag set. Andreou et al. (2013) demonstrate the value of daily financial information for macroeconomic forecasting. Here, the logic is inverted: monthly macro releases and daily financial and sentiment variables are aligned to a weekly silver-return target. MIDAS is therefore not an exotic addition; it is the appropriate linear bridge between mixed-frequency public information and the weekly EMH test.

Machine-learning models extend the alternative hypothesis from linear to nonlinear predictability. Random forests and gradient boosting can detect thresholds and interactions; LSTMs can learn sequential patterns without hand-specifying every lag. This is important because a failure of ARIMAX or VAR alone would only reject simple linear predictability, not nonlinear predictability. The deeper forecasting literature, however, urges caution. Sezer et al. (2020) review a large body of deep-learning studies in financial time series and find many reported gains, but the broader forecasting literature shows that more complex methods do not automatically beat simpler benchmarks, and that rankings often depend on the evaluation metric and test design. The thesis therefore uses machine learning not as a promise of superior performance, but as a stricter test of efficiency: if flexible nonlinear models also fail to beat the drift, the mean-efficiency conclusion becomes stronger.

3.3 Directional accuracy versus magnitude accuracy

A separate strand of the literature evaluates whether forecasts get the *direction* of returns right, rather than their magnitude. Pesaran and Timmermann (1992) provide a formal test for market-timing ability, and directional metrics remain popular in economics and finance because traders often care about whether prices rise or fall. But directional accuracy is not the same object as RMSE, MAE, or out-of-sample R^2 . A model can call slightly more signs correctly while still making large errors on the weeks that matter most; conversely, a model can reduce squared error without producing a useful trading rule. Leitch and Tanner (1991) make the broader point that conventional statistical

error measures and economic profitability can rank forecasts differently. Christoffersen and Diebold (2006) similarly treat direction-of-change forecasting as a distinct problem from point forecasting.

For an EMH thesis, this distinction is crucial. The weak- and semi-strong-form hypotheses are fundamentally about whether the conditional expected return moves away from its benchmark value. Therefore RMSE, out-of-sample R^2 , and the Diebold–Mariano test against the drift are the primary evidence. Directional accuracy is still informative, especially under the adaptive-markets interpretation: a transient sign signal may reveal episodic market structure, behavioural pressure, or regime dependence. But unless that signal improves magnitude forecasts, survives robustness checks, and is economically exploitable after costs, it should be described as directional dependence, not as a rejection of mean efficiency.

3.4 Variance predictability: GARCH, realised volatility, and HAR

The volatility literature begins from the opposite empirical fact: although returns are hard to forecast, their magnitudes are persistent. The ARCH model of Engle (1982) and the GARCH extension of Bollerslev (1986) formalised volatility clustering by allowing conditional variance to depend on past shocks and past variance. This is now one of the most established facts in financial econometrics. For this thesis, the implication is that predictable volatility does not weaken the EMH result. It strengthens the mean-variance interpretation: silver may be efficient in expected return while still risky in a forecastable way.

Realised-volatility models sharpen the target by replacing latent volatility with a model-free proxy built from higher-frequency squared returns. Barndorff-Nielsen and Shephard (2002) and Andersen et al. (2003) helped establish realised variance as a practical object for volatility measurement and forecasting. The HAR model of Corsi (2009) then became a benchmark because it captures long-memory-like volatility persistence using simple averages over short, medium, and long horizons. Patton (2011) adds an important evaluation point: because realised volatility is itself a noisy proxy for latent variance, forecast comparisons should use losses that are robust to measurement error, especially QLIKE. This justifies the thesis’s use of GARCH and HAR as volatility benchmarks and QLIKE-based Diebold–Mariano tests as the main variance-comparison evidence.

3.5 Sentiment, attention, and social media

The sentiment literature gives the thesis its most novel extension. Tetlock (2007) shows that media pessimism contains information for stock prices, while Baker and Wurgler (2006) and Barber and Odean (2008) connect investor sentiment and attention to investor behaviour. Da et al. (2011) introduce Google search volume as a direct measure of retail attention, and Antweiler and Frank (2004) show that internet message-board activity contains more reliable information about trading activity and volatility than about returns. This distinction matches the thesis’s expected result: sentiment and attention may matter more for the variance of silver returns than for the mean.

Modern NLP improves measurement. FinBERT, a BERT-based model (Devlin et al., 2019), is designed for financial language and outperforms generic methods on financial sentiment tasks (Araci, 2019), while RoBERTa and Twitter-specialised models improve contextual understanding for informal, rapidly changing online language (Liu et al., 2019; Loureiro et al., 2022). This motivates using FinBERT for financial news and Twitter-RoBERTa-style models for Reddit rather than a single generic dictionary. The measurement choice is therefore grounded in the literature: domain matters, and Reddit language is not the same as financial-news language.

The evidence on Reddit-style attention is mixed but highly relevant. Studies of Wall-StreetBets and GameStop (Lyócsa et al., 2022) show that Reddit activity can precede or accompany extreme retail-driven market episodes, but newer evidence often finds that raw activity, comment volume, and Google Trends are stronger signals than sentiment polarity alone. A focused literature search did not surface a mature, peer-reviewed silver-only equivalent; the closest evidence comes from broader equity, Reddit, and volatility studies. That absence is useful: it creates the gap this thesis fills by testing Reddit and news sentiment directly in silver, separating tone from attention, and asking whether either adds incremental value to mean or variance forecasts.

Taken together, the literature supports the thesis design. The conditional mean is the correct object for weak- and semi-strong-form efficiency tests; the drift benchmark is the correct null; nonlinear models are useful because they make the test harder to pass; directional accuracy is interesting but secondary; and volatility is expected to be more forecastable than returns. Sentiment is therefore best treated not as a guaranteed return predictor, but as an empirical ablation that may capture attention-driven volatility, especially in a market like silver where industrial demand, monetary hedging, and retail narratives can collide.

4 Data

All models are estimated on one weekly dataset, January 2015 to May 2026, on a common Friday-to-Friday (W-FRI) calendar of 591 weeks. Six data-source families are consolidated into a single feature frame (Section 4.3); the forecast target throughout is the weekly silver log-return $r_t = \log P_t - \log P_{t-1}$. Every series is observed at the Friday- $t-1$ close and lagged before use (Section 4.3):

- **Market prices** (Yahoo Finance, daily) — silver, the forecast target (COMEX front-month SI=F), and the six cross-asset benchmarks gold, copper, the US dollar index, the S&P 500, the VIX, and oil, entered as weekly log-returns (the EXOG set) (Batten et al., 2010; Zhang & Wei, 2010).
- **Rates and inflation** (FRED, daily/weekly) — the 10-year TIPS real yield, 10-year breakeven inflation, and initial jobless claims, each as a weekly change.
- **Macroeconomic** (FRED, monthly) — CPI, the federal funds rate, industrial production, and M2, entered under a point-in-time publication-availability rule.
- **Positioning** (CFTC Commitments of Traders, weekly) — managed-money and commercial net positioning on COMEX silver, as a share of open interest.
- **Sentiment** (Reddit, GDELT, NewsAPI.ai, Google Trends) — retail and news tone and attention, scored with Twitter-RoBERTa for Reddit (Loureiro et al., 2022) and FinBERT for news (Araci, 2019), averaged to a weekly mean.
- **Business sentiment (PMI)** (FRED, monthly) — a China business-confidence proxy for industrial demand, entered as a point-in-time level.

The two text-news feeds differ in coverage: GDELT is sparse—it begins only in late 2017 and has zero-article weeks even inside the test window, a documented limitation—while the paid NewsAPI.ai feed spans the full sample and is the cleaner news source; the volatility chapter (Section 7) draws its attention and sentiment signals from Reddit, paid news, and Google Trends instead. Appendix A catalogues every variable (Table 7) and details each individual series.

4.1 Technical indicators

The TECH ablation group collects momentum and trend transforms of silver’s own weekly price—MACD, RSI, rate-of-change, momentum, price-to-moving-average, Donchian position, and Bollinger %B (the full list is in Table 7)—the standard chart-analysis toolkit, included to test whether such signals carry predictive content for the return. Pure volatility and dispersion indicators are not included: they inform the conditional *variance* rather than the conditional mean a return forecast targets, so they carry little information for the return. The exact formulas and windows are tabulated in

Appendix A.2; the implementing code is in the project repository.¹

4.2 Sentiment and attention features

The volatility chapter (Section 5.3) reads each text source through two derived signals—how *much* is written (attention) and how *strong* the tone is (intensity)—rather than the signed weekly tone the return models use. For a week t with n_t items (Reddit posts or news articles) and weekly-mean tone $\bar{s}_t \in [-1, 1]$,

$$\text{attention}_t = \log(1 + n_t), \quad \text{intensity}_t = |\bar{s}_t|. \quad (9)$$

Attention is logged because weekly volume is heavily right-skewed (Reddit: median ≈ 330 , max $\approx 9,000$ posts); intensity drops the sign because for a volatility target it is the *magnitude* of sentiment, not its direction, that should move the second moment. Both are formed for Reddit (Twitter-RoBERTa tone) and the paid-news feed (FinBERT title tone) and lagged one week. Google Trends adds a third, broad-retail attention proxy: the log of Google’s search-volume index (SVI) for the query “silver”—a weekly series Google scales to 0–100 over the sample (Da et al., 2011)—so $\text{trends}_t = \log(\text{SVI}_t)$, with full 2015–2026 coverage where the news feeds are sparse.

4.3 Feature frame and split

Daily series are aggregated to a common weekly (W-FRI) grid by fixed rules: returns and FRED changes are summed over the week, sentiment is averaged, and levels (e.g. `gs_ratio_z`) and COT positioning take the week’s last value. Every feature is then lagged before it enters a model, so a value at the Friday- $t-1$ close forecasts the Friday- t return with no intra-week look-ahead—most by a uniform one week, the monthly macro block at its actual publication lag, and a best-lag ablation set at a pre-selected lag (for each predictor, the one most correlated with the target in-sample).

The split is chronological, cut on Friday boundaries: 2015–2021 for training, 2022 for validation, and 2023 to May 2026 as the held-out test window (Table 1). This places the 2024–2026 bull run in the test set by design, so the models are judged across a genuine regime shift rather than on an in-sample echo of it.

¹<https://github.com/asugar13/silver-thesis>

Table 1: Chronological train/validation/test split (W-FRI weeks).

Split	Window	Weeks	Role
Train	2015-01 – 2021-12	365	model fitting
Validation	2022	52	model & hyperparameter selection
Test	2023-01 – 2026-05	174	held-out evaluation
Total	2015 – 2026	591	

5 Methodology

5.1 Forecasting targets and benchmarks

The target for the conditional-mean analysis is the weekly silver log-return r_t of Section 2.1, forecast one week ahead. Two benchmarks frame every model. The first is a *naïve* forecast, $\hat{r}_{t+1} = r_t$, the persistence of last week’s return; it is a weak reference for a near-white-noise target and is reported only for context. The load-bearing benchmark is the *drift*, the prevailing (expanding) mean of past returns,

$$\hat{r}_{t+1}^{\text{drift}} = \frac{1}{t} \sum_{s=1}^t r_s, \quad (10)$$

which is the random-walk-with-drift forecast and is identical to ARIMA(0, 0, 0) on an expanding window. It is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008): a model that cannot beat the drift out of sample carries no exploitable conditional-mean information, so “beating the drift” is the operational form of rejecting the efficiency null hypotheses of Equations (2) and (5).

The second-moment analysis (Section 7) swaps the return for weekly *realised volatility*: realised variance is additive across days, so the weekly figure is the square root of the summed daily squared returns,

$$\text{RV}_t = \sqrt{\sum_{i \in \text{week } t} r_i^2}, \quad (11)$$

a model-free volatility proxy. Here the load-bearing floor is the *naïve* random-walk forecast $\widehat{\text{RV}}_{t+1} = \text{RV}_t$ —last week’s volatility carried forward, which has zero direction-of-change accuracy by construction—the second-moment counterpart of the drift null for the mean.

Models are evaluated by *walk-forward* one-step-ahead forecasting on the held-out 2023–2026 window (Section 4.3): at each week a model is fit on data through t and predicts $t + 1$, so no future information enters a forecast. Where the model permits, two windowing schemes are compared—an *expanding* window that grows with the

sample, and a *rolling* 100-week window that discards older data to track regime change. Re-estimation cadence is model-specific and noted below.

5.2 Conditional-mean models

The models form a deliberate ladder of increasing flexibility—from linear own-history, through linear public-information, to nonlinear machine learning—every rung judged against the same drift floor. Each exogenous predictor enters lagged one week (Section 4.3), and the ablation *ladder* adds feature groups cumulatively to a six-asset cross-return base (EXOG): the gold/silver ratio, technical indicators, monthly macro, the China PMI proxy, daily FRED indicators, COT positioning, and Reddit, GDELT, and reputable paid-news sentiment, up to an all-features rung.

ARIMA / ARIMAX. An $\text{ARIMA}(p, d, q)$ model expresses the d -th difference of the return through its own past values and past innovations,

$$\phi(L)(1-L)^d r_t = \mu + \theta(L)\varepsilon_t, \quad \phi(L) = 1 - \sum_{i=1}^p \phi_i L^i, \quad \theta(L) = 1 + \sum_{j=1}^q \theta_j L^j, \quad (12)$$

where L is the lag operator and ε_t is white noise. The order (p, d, q) is chosen by AIC on the training and validation data; the search selects $\text{ARIMA}(0, 0, 0)$ —that is, $r_t = \mu + \varepsilon_t$ —so the best own-history model coincides with the drift and gives the cleanest weak-form test. ARIMAX adds the lagged exogenous groups as regressors,

$$r_t = \mu + \beta' x_{t-1} + \varepsilon_t, \quad (13)$$

where x_{t-1} collects the lagged exogenous groups added cumulatively along the ablation ladder above (the gold/silver ratio, technical indicators, FRED daily and monthly macro, the PMI proxy, COT positioning and the sentiment feeds); this supplies the linear semi-strong test. As a complementary series-level weak-form diagnostic, the generalised spectral test of Hong and Lee (2003) checks for nonlinear own-history departures from the martingale-difference null that linear autocorrelation diagnostics may miss.

VAR. A vector autoregression of silver and the cross-asset returns, stacked in a vector y_t , with lag order p chosen by AIC,

$$y_t = c + \sum_{i=1}^p A_i y_{t-i} + \varepsilon_t, \quad (14)$$

which captures linear feedback among the assets; Granger-causality and impulse-response diagnostics accompany it descriptively in the project repository.

MIDAS. Mixed-data-sampling regressions keep a predictor’s recent high-frequency

path instead of collapsing it to a single weekly number, and learn how much each lag should matter. A high-frequency predictor’s K lags are compressed into one term through a normalised lag-weight curve,

$$m_{j,t}(\theta_j) = \sum_{k=1}^K w_j(k; \theta_j) x_{j,t-k}, \quad \sum_{k=1}^K w_j(k; \theta_j) = 1, \quad (15)$$

where the weights follow the Beta or exponential-Almon family (Ghysels et al., 2004, 2007). The weekly forecast then regresses the silver return on its own autoregressive lags and the compressed mixed-frequency blocks,

$$r_t = \alpha + \sum_{i=1}^p \phi_i r_{t-i} + \sum_j \beta_j m_{j,t}^{(d)}(\theta_j) + \sum_m \delta_m m_{m,t}^{(M)}(\eta_m) + \sum_q \gamma_q q_{q,t-1} + \varepsilon_t, \quad (16)$$

where $m^{(d)}$ are daily MIDAS blocks, $m^{(M)}$ publication-lagged monthly-macro blocks, and $q_{q,t-1}$ weekly-release controls (jobless claims, COT). The shape parameters are estimated by nonlinear least squares (the linear coefficients profiled out), with the weight family selected on the validation window. The payoff is parsimony: rather than $K = 20$ free coefficients per daily predictor, two weight parameters trace a smooth lag profile, which is what makes daily information usable in a weekly forecast without overfitting. Two variants are run—monthly macro at three lags, and a multi-frequency model adding the six cross-asset returns at 20 daily lags (Andreou et al., 2013).

Random forest and gradient boosting. Nonlinear ensembles that admit thresholds and interactions among the lagged features—a random forest (Breiman, 2001a) and gradient-boosted trees, XGBoost (Chen & Guestrin, 2016). Hyperparameters are grid-searched per ablation rung on the training-plus-validation data using time-series cross-validation—contiguous expanding folds that always validate on weeks *after* those used to fit, so no future information leaks into tuning—and the model is re-fit every four weeks in the walk-forward.

LSTM. A long short-term memory recurrent network (Hochreiter & Schmidhuber, 1997) that reads a fixed window of recent weeks and learns its own representation of the past; a single recurrent layer with dropout and early stopping is used, given the small sample (a few hundred training sequences). Because a single trained network is one random draw (weight initialisation and batch order), every variant is trained over five seeds: reported metrics are seed-means, and the median-RMSE seed supplies the forecast—a representative draw rather than an ensemble average, which would understate the error. A weekly-input model and a daily-input variant (a 60-trading-day window) are estimated, the latter testing whether daily-resolution information helps forecast the weekly return.

The classical and tree models are run under both windowing schemes; the LSTM uses a single expanding-window training pass, since a 100-week rolling window leaves too few sequences to train stably. In all, the conditional-mean analysis spans six model families—ARIMA/ARIMAX, VAR, MIDAS, the random forest, XGBoost, and the LSTM—plus two daily-input variants (for MIDAS and LSTM) that read the higher-frequency data directly, for eight specifications judged against the common drift floor.

5.3 Conditional-variance models

The second-moment analysis models weekly realised volatility (RV_t , Eq. 11 in Section 5.1). Its predictors are the heterogeneous-autoregressive (HAR) lags—1-week, monthly (4-week mean), and quarterly (12-week mean) trailing realised volatility—the three horizons that, in Corsi’s heterogeneous-market account (Corsi, 2009), proxy short-, medium-, and long-horizon traders and together reproduce volatility’s long-memory persistence parsimoniously. They are optionally extended with the six cross-asset realised volatilities, Reddit attention and sentiment-intensity measures, paid-news attention and intensity, and Google Trends search interest; these added blocks form the volatility chapter’s HAR-X ablation rungs, and Google Trends enters here only, not in the conditional-mean models.

GARCH. A GARCH(1,1) model (Bollerslev, 1986; Engle, 1982), in which the return is a constant mean plus a conditionally heteroscedastic shock, $\varepsilon_t = \sigma_t z_t$, whose variance follows

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (17)$$

It is fit by maximum likelihood and re-estimated walk-forward—the parametric benchmark for conditional variance.

HAR-RV and HAR-X. The heterogeneous autoregressive model of Corsi (2009) regresses next-week realised volatility on its own trailing averages over short, medium, and long horizons; HAR-X augments that base with an exogenous block z_{t-1} ,

$$RV_t = \underbrace{\beta_0 + \beta_w RV_{t-1}^{(w)} + \beta_m RV_{t-1}^{(m)} + \beta_q RV_{t-1}^{(q)}}_{\text{HAR-RV}} + \gamma' z_{t-1} + \varepsilon_t, \quad (18)$$

where $RV^{(w)}$, $RV^{(m)}$, $RV^{(q)}$ are the trailing 1-week, 4-week, and 12-week mean realised volatilities, and z_{t-1} is the added predictor block—the six cross-asset realised volatilities, Reddit attention and intensity, paid-news attention and intensity, or Google Trends, entered as separate ablation rungs ($z_{t-1} = \emptyset$ recovers HAR-RV). Both are estimated by OLS.

Tree ensembles. Random forest and XGBoost on the HAR-plus-cross-asset feature

set, isolating any nonlinear gain over the linear HAR-X.

5.4 Evaluation protocol and significance tests

All models are scored on the same 2023–2026 test window of n weeks; write r_t for the realised return and $\hat{r}_t$ for the forecast. Point-forecast accuracy is measured by root-mean-squared and mean-absolute error,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (r_t - \hat{r}_t)^2}, \quad \text{MAE} = \frac{1}{n} \sum_{t=1}^n |r_t - \hat{r}_t|, \quad (19)$$

the primary metrics, since efficiency is a conditional-mean (magnitude) claim. The directional lens is directional accuracy and its magnitude-weighted form,

$$\text{DA} = \frac{1}{n} \sum_{t=1}^n \mathbf{1}[\text{sign } r_t = \text{sign } \hat{r}_t], \quad \text{WDA} = \frac{\sum_t |r_t| \mathbf{1}[\text{sign } r_t = \text{sign } \hat{r}_t]}{\sum_t |r_t|}, \quad (20)$$

WDA weighting each correct call by the size of the move, so a missed large week costs more than a missed flat one. Because the volatility target is non-negative, DA and WDA do not apply to it; the volatility chapter instead scores direction with direction-of-change accuracy (DCA)—whether the model correctly called the rise or fall in log RV. Effect size relative to the drift is the Campbell–Thompson out-of-sample R^2 (Campbell & Thompson, 2008),

$$R_{\text{OS}}^2 = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_t)^2}, \quad (21)$$

with $\bar{r}_t$ the prevailing-mean drift forecast, so $R_{\text{OS}}^2 > 0$ means the model beats the random walk out of sample.

Significance rests on the following tests. The **Diebold–Mariano** test of equal predictive accuracy (Diebold & Mariano, 1995), with a lag-1 Newey–West (Newey & West, 1987) long-run variance, compares each model to its floor—the drift for returns, the naïve RV_{t-1} random walk for volatility. For returns it is reported under squared-error loss (the headline) and absolute-error loss (a heavy-tail robustness check), and for volatility under the QLIKE loss (Patton, 2011)—robust, like squared error, to the noise in the realised-volatility proxy, but far better powered than squared-error DM on heavy-tailed RV, where a handful of extreme weeks swamp the loss differential. The **Pesaran–Timmermann** test (Pesaran & Timmermann, 1992) is the matching significance test for the directional (DA/WDA) reading.

Two robustness layers accompany the headline tests: a breakdown by calendar sub-period across 2023–2026, to confirm that performance is not driven by a single regime, and a re-run of the full battery with the exceptional 2025 bull run excluded. The load-bearing verdict throughout is the Diebold–Mariano test against the relevant floor—the

drift for returns, the naïve RV_{t-1} random walk for volatility; the directional results are a secondary, economically motivated lens.

6 The Conditional Mean: Forecasting Weekly Returns

This chapter evaluates the conditional-mean models of Section 5.2 on the held-out 2023–2026 window, taking the weak- and semi-strong-form questions in turn. The benchmark throughout is the prevailing-mean drift, and the load-bearing test is the Diebold–Mariano (DM) comparison against it; the directional metrics are a secondary lens. Table 2 collects the headline numbers.

6.1 Weak form: own-history models

At the series level, the own-history evidence already points to weak-form efficiency. A Ljung–Box test finds only marginal linear autocorrelation in weekly returns ($p \approx 0.06$ at lag 10), and the generalised spectral test of Hong and Lee (2003) —which probes nonlinear as well as linear departures—*fails to reject* the martingale-difference null at every lag order tried (bootstrap $p \approx 0.90$ – 0.95 for $p \in \{5, 10, 20\}$). The conditional mean is therefore unpredictable from past returns against both linear and nonlinear alternatives, while the same diagnostics strongly reject independence in the *squared* returns ($p \approx 0$)—the predictable structure is confined to the variance, not the mean.

The forecasting models confirm this. The AIC search over own-history ARIMA orders selects ARIMA(0, 0, 0) outright, so the best linear own-history forecast *is* the drift and there is nothing left to test. The only own-history variant that departs from it, a 100-week rolling ARIMA, does not significantly beat it—a marginal, insignificant edge that comes from tilting toward the 2025 trend and reverses once that year is removed. Past returns, linear or nonlinear, carry no exploitable signal for the conditional mean.

6.2 Semi-strong form: public information

Expanding the information set does not change the verdict. Table 2 reports each family’s RMSE-best variant over the full ablation ladder—cross-asset returns, the gold/silver ratio, monthly macro and the China PMI proxy, daily FRED indicators, COT positioning, and Reddit, GDELT and paid-news sentiment—with the tree ensembles and the LSTM free to combine them nonlinearly. The spread is strikingly tight: every family lands within about one percent of the drift’s 5.19% weekly RMSE, and the best of them—an XGBoost on the cross-asset returns plus paid-news sentiment—improves

on the drift by just 0.05 points ($R_{OS}^2 = +1.8\%$). Figure 3 shows the same picture: the bars sit on the drift floor.

Table 2: Weekly return forecasts—best variant per model family (RMSE-selected), full 2023–2026 test window. RMSE and MAE are in percentage points per week; R_{OS}^2 is the Campbell–Thompson out-of-sample R^2 versus the drift (%). Walk-forward window qualifiers (expanding / rolling) are omitted from the variant labels. **No model row beats the drift floor on a DM test** (smallest $p = 0.43$, squared-error loss; unchanged under absolute error). The drift is the random-walk-with-drift benchmark (= ARIMA(0, 0, 0)); its WDA doubles as the always-up directional rate.

Family	Best variant	RMSE	MAE	R_{OS}^2	DA	WDA
XGBoost	EXOG+ALLPaid	5.14	3.69	+1.84	0.540	0.574
MIDAS (daily)	EXOG-d+GS_HF	5.14	3.68	+1.78	0.586	0.607
LSTM (daily)	EXOG-GS-SentimentPaid	5.16	3.70	+1.27	0.561	0.568
Random Forest	EXOG+News	5.16	3.72	+0.98	0.517	0.560
LSTM	EXOG-ALLPaid	5.17	3.71	+0.59	0.556	0.563
VAR	EXOG+GS+Sentiment	5.18	3.66	+0.41	0.603	0.594
ARIMAX	ARIMAX+GS+Sentiment	5.18	3.65	+0.23	0.609	0.588
MIDAS	EXOG	5.21	3.71	−0.68	0.546	0.507
<i>Drift (prevailing mean)</i>	—	5.19	3.71	0.00	0.557	0.588
<i>Naïve (t−1)</i>	—	7.59	5.40	−112.9	0.491	0.490

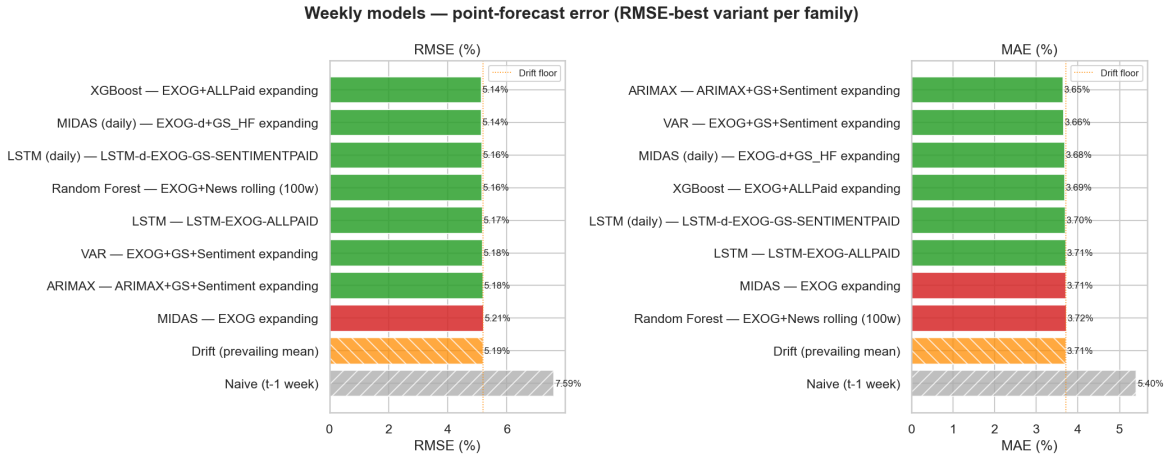


Figure 3: Point-forecast error of each family’s RMSE-best variant against the drift floor (dotted), full test window. Green bars fall below the drift’s RMSE/MAE, red above; all sit within a hair of it, and the gap is statistically insignificant for every family. The naïve $t - 1$ benchmark (hatched) is far worse—persistence is a poor reference for a return.

None of this is statistically distinguishable from the random walk. Across all eight families and both selection criteria—sixteen family-selections in total—not one beats the drift on a DM test (squared-error loss; the smallest p is 0.43), and the verdict is unchanged under absolute-error loss, the heavy-tail robustness check. The Campbell–Thompson R_{OS}^2 tells the effect-size version of the same story: it spans +1.8% to −0.7% for the magnitude-best variants and turns sharply negative for the directionally-selected

variants, so on average a public-information model forecasts the weekly return no better than its own expanding historical mean. The null hypothesis cannot be rejected for any model class—the linear ARIMAX, VAR and MIDAS regressions, the tree ensembles, and the recurrent network all collapse onto the drift. That model classes with such different inductive biases—a linear regression, a tree ensemble, a recurrent network—converge on the same drift-equivalent forecast is a benign instance of the model multiplicity Breiman (2001b) describes: when the conditional mean carries no signal, the choice among well-fit model forms is immaterial. Every feature in the ladder—cross-asset returns, positioning, macro, and sentiment alike—was public information at the forecast date, so the whole ladder is one cumulative test of semi-strong-form efficiency, and none of it beats the drift: the semi-strong null is not rejected.

6.3 Direction versus magnitude

A weaker, direction-only signal does surface. Several families call the sign of the weekly return more often than the drift does: because the prevailing mean of past returns is positive, the drift always forecasts an up week, so its directional hit rate is merely the share of weeks that rose. By the directionally-selected (WDA-best) criterion the linear models clear that bar— $DA \approx 0.62$ (Figure 4)—and a Pesaran–Timmermann (PT) test confirms genuine sign-timing in the linear ARIMAX and VAR ($p = 0.004$ and 0.005) and in the daily-input MIDAS. But the signal is magnitude-worthless. The variants that maximise directional accuracy do so at the expense of point-forecast error: they carry negative R_{OS}^2 (VAR’s WDA-best loses 14% to the drift) and win no DM test. It is also fragile—concentrated in the linear models and the post-hoc best-selected variants, while across the full ablation ladders of the nonlinear families it appears at no more than the chance rate (one of 38 random-forest variants, none of 38 for XGBoost). We read it as consistent with the adaptive-markets view of Lo (2004)—a weak, episodic signal, not evidence against efficiency in the conditional mean.

The adaptive-markets reading can be explored more directly, but the evidence distinguishes temporal instability from volatility-state dependence. The diagnostic retains the two one-week-lagged predictors responsible for the linear models’ directional result: COT managed-money net positioning, whose addition made the ARIMAX ladder PT-significant, and the paid-news title sentiment score, the only sentiment measure whose PT result replicated in both VAR and MIDAS. Pooling them gives a deliberately parsimonious specification for tracing the previously detected signal through time, rather than a new search across the full feature set.

A rolling 52-week Pesaran–Timmermann test of its expanding-window directional forecasts is insignificant when pooled over the sample ($p = 0.145$). Of 433 rolling

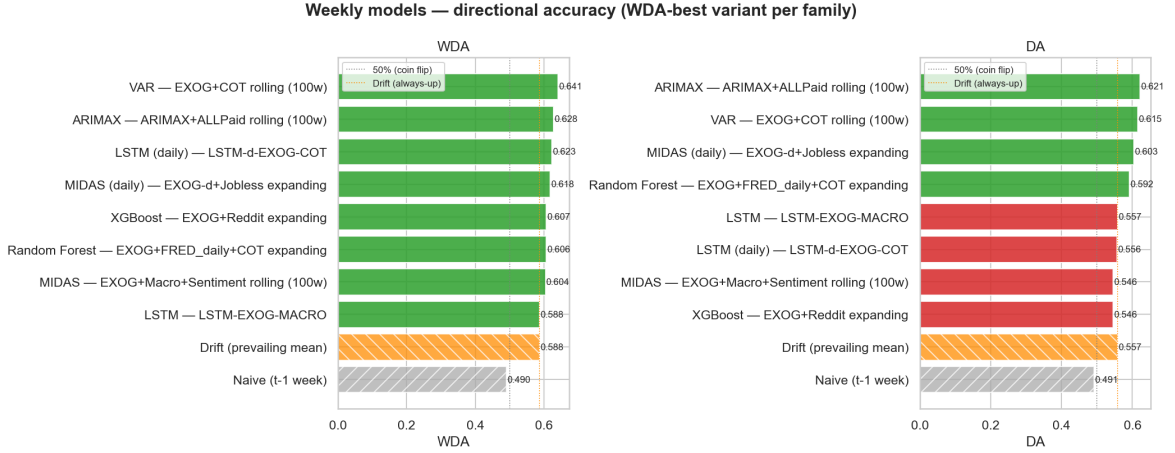


Figure 4: Directional accuracy of each family’s WDA-best variant. The drift bar *is* the always-up line (its sign is constant-positive); green beats it. Several families clear it, but the directionally-selected variants are the ones with the worst RMSE (Table 2), so the directional edge does not translate into magnitude accuracy.

windows, 26 (6.0%) produce $p < 0.05$, with their endpoints concentrated in 2021–22 and 2024–25. These are not four independent annual rejections: adjacent windows share 51 of 52 observations. Moreover, examining 433 p -values means that some will fall below 0.05 by chance even if no genuine signal exists, and no correction is applied for making so many comparisons. The clustering is therefore descriptive evidence of temporal instability, not a standalone rejection of constant efficiency. Table 3 shows both the recurrence and its limited extent.

Table 3: Rolling directional significance by window-ending year. Each row summarises the weekly rolling 52-week PT tests whose final observation falls in that calendar year; it is not a separate annual PT test. “Minimum p ” is the smallest rolling-window p -value observed during the year. Adjacent windows share 51 observations, so the final column counts overlapping endpoints rather than independent rejections.

Endpoint year	Rolling windows	Minimum p	Windows with $p < 0.05$
2018	50	0.161	0
2019	52	0.166	0
2020	52	0.179	0
2021	53	0.009	7
2022	52	0.021	3
2023	52	0.145	0
2024	52	0.024	12
2025	52	0.033	4
2026	18	0.269	0

A two-regime Markov-switching regression asks the separate question of whether that instability is associated with volatility states. It identifies a long *calm* regime (expected duration ≈ 71 weeks) and short *turbulent* spells (≈ 12 weeks), with estimated variances

of 10.47 and 64.86; equality of the two variances is rejected (Wald $p = 0.003$). The smoothed turbulent-state probability also rises around large absolute weekly returns (Appendix Figure 8), supporting the high-variance interpretation. However, none of the 26 significant rolling-test endpoints lies in the turbulent state, and only seven of their underlying 52-week windows contain even one turbulent observation. The paid-news coefficient is more negative in the turbulent state in point-estimate terms (-2.23 versus -0.22), but the cross-regime difference is not significant (Wald $p = 0.116$). The volatility regimes are therefore well distinguished, but they do not explain the episodic directional result (Figure 5). Both diagnostics leave the magnitude-efficiency verdict unchanged.

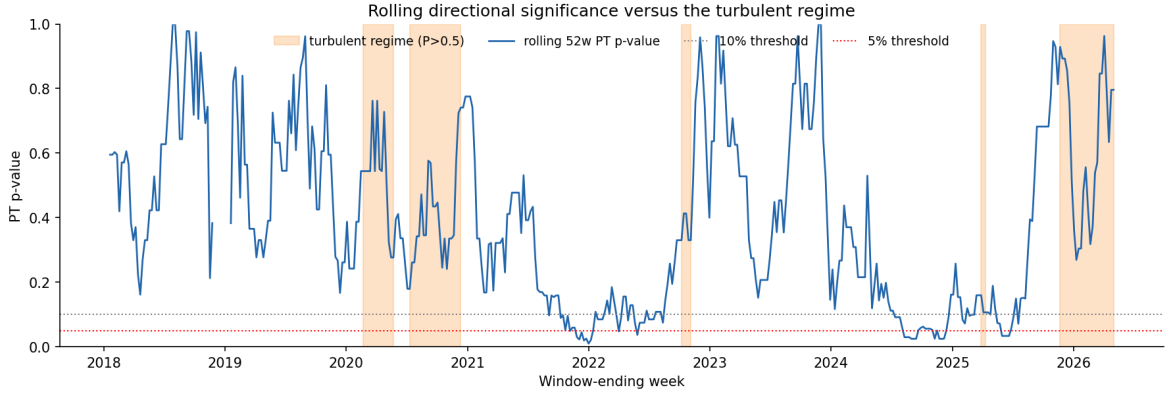


Figure 5: Temporal directional instability versus volatility regimes. The line is the rolling 52-week Pesaran–Timmermann p -value of the COT + paid-news directional forecast; orange shading marks weeks where a two-regime Markov-switching model places the high-variance *turbulent* state. The low- p clusters and turbulent shading largely do not coincide: the volatility classification does not explain the timing of directional significance.

6.4 Robustness

The conclusion strengthens under scrutiny. Re-running the full battery with the exceptional 2025 bull removed—pooling 2023, 2024 and 2026, 122 weeks—again yields zero of sixteen family-selections beating the floor; if anything the directionally-selected variants now fall significantly *behind* it (VAR $p = 0.048$, weekly MIDAS $p = 0.029$). The per-period view (Figure 6) makes the point visually: with one exception, no family holds a positive out-of-sample R^2 across all four sub-periods—a good year is offset by a bad one, the signature of regime luck rather than skill—and the high directional accuracy of 2025 is simply the bull run rewarding the always-up tilt. The exception, XGBoost, stays positive in every sub-period but by margins ($+0.3$ to $+5.8\%$) that pool to the insignificant full-window result (DM $p = 0.43$): the luckiest draw among sixteen selections, not evidence of skill. The expanding and 100-week rolling windows tell the same story, with the rolling variants the worse of the two on balance. Weekly silver returns are, on this evidence, unforecastable from public information.

WDA by sub-period (WDA-best variants)					OOS R^2 (%) vs Drift by sub-period (RMSE-best variants)				
ARIMAX	0.761	0.493	0.725	0.492	ARIMAX	+5.4	+7.3	-3.3	-2.5
VAR	0.677	0.561	0.793	0.488	VAR	+5.2	+6.7	-3.9	-1.4
Random Forest	0.570	0.640	0.670	0.528	Random Forest	-4.7	-3.4	+3.2	+3.8
XGBoost	0.537	0.613	0.629	0.654	XGBoost	+0.3	+5.8	+2.5	+0.5
LSTM	0.492	0.570	0.757	0.498	LSTM	-0.9	+4.1	-0.3	+1.0
MIDAS	0.506	0.585	0.714	0.597	MIDAS	+0.5	+2.0	-3.7	-0.1
MIDAS (daily)	0.589	0.714	0.663	0.487	MIDAS (daily)	-0.7	+12.4	-8.0	+5.5
LSTM (daily)	0.493	0.666	0.719	0.627	LSTM (daily)	+1.7	+2.1	-1.0	+2.2
Drift (always-up)	0.492	0.570	0.757	0.498					
	2023 (choppy)	2024 (bull start)	2025 (bull run)	2026 (YTD)		2023 (choppy)	2024 (bull start)	2025 (bull run)	2026 (YTD)

Figure 6: Per-period robustness. *Left*: directional accuracy (WDA) by calendar year for the WDA-best variants—uniformly high in the 2025 bull, much weaker elsewhere. *Right*: out-of-sample R^2 versus the drift by year for the RMSE-best variants; the sign flips across periods for every family except XGBoost, whose uniformly positive margins are thin and never significant, so no stable magnitude edge emerges.

7 The Conditional Variance: Forecasting Weekly Volatility

7.1 Realised volatility is forecastable

Where the conditional mean was unforecastable, the conditional variance is not. Every model in the chapter—GARCH(1,1), the linear HAR-RV, and the two tree ensembles on the HAR lags—beats the naïve RV_{t-1} floor out of sample (Table 4). HAR-RV cuts the naïve mean-squared error by roughly a third ($R^2_{OS} = 0.29$ against the RV mean) and calls the direction of the weekly volatility change in 71% of test weeks; GARCH and the trees land in the same band ($R^2_{OS} = 0.26$ – 0.32 , DCA 0.68–0.71), while the naïve floor’s R^2_{OS} is negative (and its direction-of-change accuracy zero by construction). Figure 9 (Appendix B) shows the HAR forecast tracking the realised series across the full window—calm bands and run-ups alike.

Significance depends on the loss. Under the proxy-robust QLIKE loss every model significantly beats the naïve floor (Diebold–Mariano $p < 0.01$ throughout); under squared-error loss the same comparisons are ties ($p \approx 0.2$ – 0.3). This is not a contradiction but a power problem: realised volatility is heavy-tailed, so a handful of extreme weeks—chiefly the late-January 2026 spike—dominate the squared-error differential and inflate its variance, which is why QLIKE is the headline test (Section 5.4). The consistent reading is that silver’s weekly volatility carries a strong, persistent, statistically detectable own-history signal—in sharp contrast to its return.

Table 4: Weekly realised-volatility forecasts—headline models, full 2023–2026 test window (174 weeks). RMSE and MAE are in realised-volatility points (percent, i.e. the raw RV scaled by 100). R^2_{OS} is the out-of-sample R^2 versus the unconditional RV mean (so the naïve RV_{t-1} floor itself scores -0.09); DCA is direction-of-change accuracy on $\Delta \log \text{RV}$. **Every model beats the naïve floor under the proxy-robust QLIKE loss** (Diebold–Mariano $p < 0.01$ throughout); under squared-error loss the same gaps are ties ($p \approx 0.2$ – 0.3), the heavy-tail power problem of Section 5.4. The naïve floor’s negative R^2_{OS} and zero DCA are by construction.

Model	RMSE	MAE	R^2_{OS}	DCA
XGBoost (HAR)	3.13	1.76	0.32	0.690
Random Forest (HAR)	3.15	1.64	0.32	0.713
HAR-RV	3.20	1.63	0.29	0.707
GARCH(1,1)	3.28	1.80	0.26	0.678
<i>Naïve</i> (RV_{t-1})	3.99	2.01	-0.09	0.00

7.2 Cross-asset spillover and nonlinearity

What does *not* help is as informative as what does. Augmenting HAR-RV with the six cross-asset realised volatilities—the volatility-spillover hypothesis—makes the linear model significantly *worse*: HAR+EXOG raises RMSE from 3.20 to 3.27 percent and loses to bare HAR under both QLIKE and squared-error loss ($p \approx 0.03$; Table 5). At the weekly horizon the cross-asset block adds estimation noise, not signal; silver’s forecastable variance is its own, not a spillover from gold, copper, or the broad risk complex.

Nonlinearity does not rescue it either. Random forest and XGBoost on the HAR lags sit in the same RMSE band as linear HAR-RV, and the same trees on HAR+EXOG are no better than on HAR alone (Diebold–Mariano ties)—the cross-asset block is dry for the trees too. What *does* matter is the multi-horizon structure itself: collapsing HAR to a single AR(1) lag significantly degrades the random forest ($p = 0.001$), confirming that the forecastable signal is the short/medium/long volatility persistence Corsi’s HAR is built to capture. The parsimonious HAR is therefore the chapter’s right benchmark, and the attention and sentiment block of the next section is the only augmentation that earns its place.

7.3 Sentiment and attention

The sentiment and attention ablation (Table 5) asks an incremental question: does adding a public-information block improve on the same estimator (HAR, the random forest, or XGBoost) using only its HAR realised-volatility lags? The answer is clearest for the linear HAR specification. Reddit sentiment, combined Reddit attention/sentiment, paid-news attention, and the combined paid-news rung all significantly beat HAR-RV

Table 5: HAR-X attention/sentiment ablation on weekly realised volatility. Each rung is fitted and scored on the common sample (174 weeks where all attention/sentiment features exist) and Diebold–Mariano-tested against its family’s HAR baseline, so the baseline levels differ slightly from the headline Table 4. RMSE is in percent ($\times 100$); R^2_{OS} is versus the RV mean; DCA is direction-of-change accuracy on $\Delta \log RV$. The two right columns are DM statistics under the proxy-robust QLIKE loss (primary) and squared-error loss (MSE); **a negative value favours the augmented rung** over bare HAR. Significance: $^\dagger p < 0.10$, $^* p < 0.05$, $^{**} p < 0.01$, $^{***} p < 0.001$.

Variant	RMSE	R^2_{OS}	DCA	QLIKE-DM	MSE-DM
<i>Linear HAR-RV</i>					
HAR (baseline)	3.205	0.293	0.707	—	—
+EXOG (spillover)	3.268	0.264	0.718	+2.23*	+2.16*
+Reddit attention	3.191	0.299	0.707	−2.60**	−1.92 †
+Reddit sentiment	3.195	0.297	0.713	−3.36***	−2.61**
+Reddit (both)	3.191	0.299	0.718	−3.01**	−2.20*
+Google Trends	3.156	0.314	0.701	−0.01	−1.90 †
+Paid attention	3.172	0.307	0.718	−3.33***	−2.41*
+Paid sentiment	3.205	0.293	0.701	+1.44	+0.06
+Paid (both)	3.164	0.311	0.718	−2.68**	−2.17*
<i>Random forest</i>					
AR(1) only	3.299	0.251	0.661	+3.24**	+1.84 †
HAR (baseline)	3.168	0.309	0.707	—	—
+EXOG (spillover)	3.377	0.215	0.684	+1.53	+1.64
+Reddit attention	3.249	0.273	0.695	+1.97*	+0.98
+Reddit sentiment	3.189	0.300	0.695	+0.75	+0.88
+Reddit (both)	3.179	0.304	0.701	+1.72 †	+0.43
+Google Trends	3.147	0.318	0.707	−0.78	−0.27
+Paid attention	3.133	0.324	0.684	+0.23	−0.87
+Paid sentiment	3.130	0.325	0.713	−1.54	−1.56
+Paid (both)	3.102	0.337	0.684	−0.34	−1.37
<i>XGBoost</i>					
AR(1) only	3.429	0.190	0.632	+0.25	+0.64
HAR (baseline)	3.329	0.237	0.661	—	—
+EXOG (spillover)	3.419	0.195	0.684	+0.60	+0.46
+Reddit attention	3.099	0.339	0.684	−2.36*	−2.07*
+Reddit sentiment	3.194	0.298	0.707	−2.44*	−1.83 †
+Reddit (both)	3.170	0.308	0.707	−0.74	−1.29
+Google Trends	3.255	0.270	0.667	−1.83 †	−0.76
+Paid attention	3.205	0.292	0.695	−2.20*	−1.88 †
+Paid sentiment	3.090	0.342	0.678	−2.29*	−1.53
+Paid (both)	3.210	0.290	0.684	−1.68 †	−1.20

under QLIKE—and most under squared-error too—while Google Trends and paid-news sentiment alone do not. Trends is the telling case: it posts the *lowest* RMSE of any linear rung yet ties bare HAR on QLIKE ($p \approx 0.99$), the clean RMSE-versus-QLIKE divergence the heavy-tailed loss predicts. The effect is therefore not a wholesale replacement of the HAR volatility structure; it is a modest but statistically visible increment on top of volatility persistence.

The tree-based results do not overturn that conclusion, but they make it more conditional. Random forests record slightly better point errors for some augmented specifications, yet the Diebold–Mariano tests mostly classify those gains as ties against the RF(HAR) baseline. XGBoost gives the strongest nonlinear evidence: the Reddit-attention rung significantly improves both MSE and QLIKE relative to XGB(HAR), and several sentiment/attention variants improve one loss even when the other remains a tie. Thus the public-information signal is useful for the linear HAR, selectively useful for XGBoost, and largely inert for the random forest—a model-dependent increment, not a uniform one. Overall, HAR remains the central and most stable benchmark, while sentiment and attention add small, loss-dependent forecasting gains rather than changing the model ranking outright.

7.4 Economic significance: a Value-at-Risk backtest

A QLIKE improvement is a statistical statement; risk management asks an economic one—does the better volatility forecast actually give a better risk number? We test it directly: each weekly RV forecast $\hat{\sigma}_t$ becomes a one-week Value-at-Risk for the silver return, $\text{VaR}_t(\alpha) = \mu + \hat{\sigma}_t q_\alpha$, and we ask how well it brackets the realised weekly losses over the test window. Only $\hat{\sigma}_t$ differs across the rows of Table 6, so any calibration difference is the forecast’s doing.

Table 6: One-week VaR backtest, 174-week test window (Normal innovations; a Student- t_5 version leaves the ranking unchanged). *Breach* is the realised breach rate (it should sit near the target α); *pinball* is the mean quantile loss ($\times 10^4$, lower is better). Every forecast-based model beats the naïve floor on the pinball loss (5% tail, Diebold–Mariano: HAR $p < 0.001$, RF and XGBoost $p \leq 0.004$); adding sentiment leaves these numbers essentially unchanged.

Model	5% VaR		1% VaR	
	breach	pinball	breach	pinball
Naïve (RV_{t-1})	8.0%	66.3	4.0%	24.1
GARCH(1,1)	3.4%	58.5	1.7%	20.8
HAR	6.3%	59.6	2.9%	21.2
HAR + Reddit attention	5.7%	59.6	2.3%	20.8
RF (HAR)	4.6%	56.2	1.7%	19.2
XGBoost (HAR)	3.4%	57.0	2.3%	20.6

The result is clear-cut, and it holds in every model class. The HAR, random-forest and XGBoost VaRs are all well-calibrated and beat the naïve floor on the quantile (pinball) loss (Diebold–Mariano $p \leq 0.004$), whereas the naïve RV_{t-1} VaR over-breaches and is poorly calibrated in the tail (Figure 7); the smoother random-forest forecast is, if anything, the best-calibrated row. Forecastable volatility is therefore not merely detectable but economically useful—the difference between a usable and an unusable tail-risk measure. The sentiment extension is shown on HAR because that is where attention improves the point forecast most clearly (Section 7.3); even there it barely moves the numbers (Table 6)—the VaR gain belongs to the volatility model, not to sentiment. The improvement sentiment brings registers only when averaged over the whole sample, while a VaR is decided by a few extreme weeks in the tail—too few for that small gain to matter.

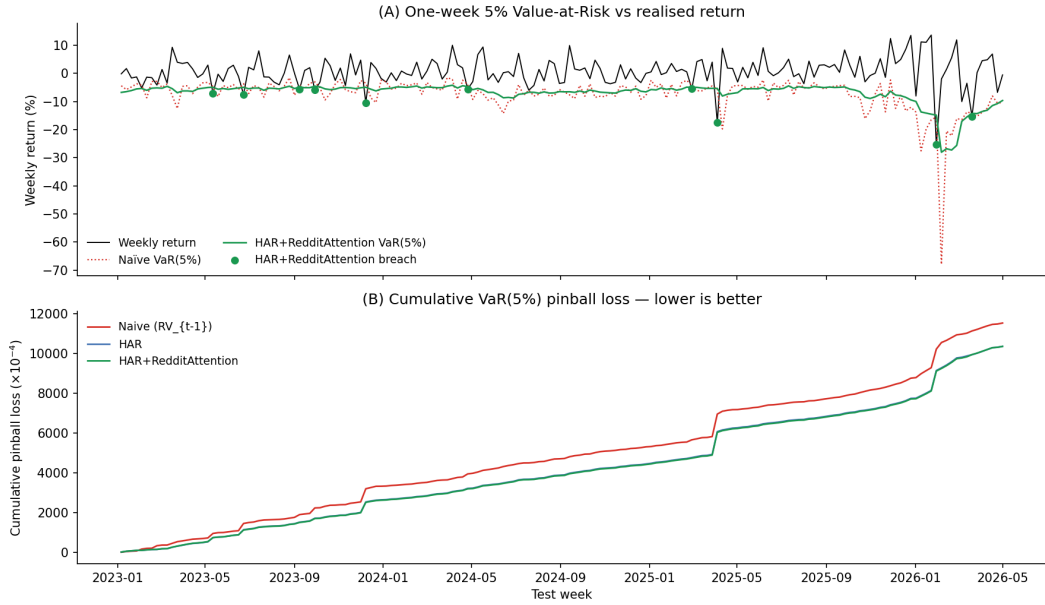


Figure 7: (A) The one-week 5% VaR from the HAR forecast (green) against the realised return; the naïve RV_{t-1} VaR (red dotted) is jagged and over-reacts to the last week, overshooting badly around the January 2026 spike. (B) Cumulative VaR(5%) pinball loss: the naïve floor (red) sits clearly above the HAR forecast (green), the gap opening at each volatility burst. Forecastable volatility converts into materially better tail risk.

This closes the loop the title opens. The conditional mean is unforecastable and the conditional variance is forecastable, and the backtest shows the forecastable moment is the one a risk system actually consumes: a forecast-based VaR is materially better calibrated than the random-walk benchmark a manager would otherwise default to.

8 Discussion

The two results chapters are not two studies but one decomposition seen in full. The conditional-moment factorisation of Section 2 splits the weekly return into a predictable mean and a time-varying variance, $r_t = \mu_t + \sigma_t z_t$, and the thesis asked whether each moment is forecastable from its own past and public information under one benchmark protocol. The verdict is sharp and opposite for the two. The conditional mean (Section 6) is not forecastable: the own-history null survives—the AIC search selects the drift itself and the generalised spectral test fails to reject the martingale-difference property—and so does the semi-strong null, with no model in the eight family ladder beating the drift floor out of sample. The conditional variance (Section 7) plainly is forecastable: HAR, GARCH and the tree ensembles all clear the naïve realised-volatility floor by a third of its mean-squared error. “The mean is a martingale, the variance is not” is therefore a single empirical statement about one series—not a failure on the mean redeemed by a success on the variance.

That pairing is fully consistent with market efficiency rather than in tension with it. The efficient-market hypothesis (Fama, 1970, 1991) constrains the conditional mean—the risk-adjusted expected return—and says nothing about the conditional variance. Forecastable volatility is not a tradable arbitrage: knowing that next week is likely to be turbulent does not tell an investor which way the price will move, and an efficient market processing information in bursts should in fact display exactly the volatility clustering that HAR exploits. The variance result thus strengthens rather than undercuts the efficiency reading: the one moment that carries genuine, persistent structure is precisely the one efficiency leaves unconstrained, and the moment efficiency does constrain is the one we cannot forecast.

The only crack in the mean’s efficiency is directional, weak, and best read through an adaptive-markets lens. A sign-timing signal surfaces in the linear models—Pesaran–Timmermann significance from COT positioning and the paid-news title score in ARIMAX and VAR—and the 100-week rolling ARIMA shows a transient edge over the expanding drift; but both are magnitude-worthless (they win no Diebold–Mariano test and carry negative out-of-sample R^2), concentrated in the 2025 bull, and non-replicating across the nonlinear families. The separate rolling test likewise produces low p -values only in clusters, although consecutive tests reuse 51 of 52 observations and inspecting hundreds of p -values makes some chance crossings of the 0.05 threshold likely; the pattern is descriptive. This temporal instability is consistent with the adaptive-markets hypothesis of Lo (2004), under which efficiency tightens and loosens with conditions, but it does not identify those conditions. In particular, the Markov analysis detects credible calm and turbulent variance states yet finds no alignment between turbulence and the

significant directional windows, while the paid-news coefficients do not differ significantly across states (Figure 5). We therefore retain these directional traces as a cautious, unexploitable adaptive-efficiency caveat, not as evidence of volatility-regime-bound predictability or against mean efficiency.

The sentiment evidence sharpens the same mean-versus-variance contrast and illustrates the gap between statistical and economic significance. Tested as an explicit ablation on both moments, public attention and sentiment never *significantly* improved the mean—no rung beat the drift on a Diebold–Mariano test—even though some sentiment feed entered six of the eight families’ RMSE-best variants; that recurring appearance stays within noise, so it reinforces rather than dents the efficiency verdict. For the variance the verdict flips. Measured against the HAR baseline (each model judged against itself on HAR lags only), Reddit and paid-news attention cut the linear HAR’s QLIKE loss at $p < 0.01$ —under the proxy-robust loss that is the chapter’s primary metric for heavy-tailed volatility—a robust improvement consistent with the media-sentiment effects of Tetlock (2007) and the sentiment-augmented volatility forecasts of Audrino et al. (2020). The gain is small in magnitude—it sharpens an already-strong HAR rather than transforming it—but statistically solid under the loss that matters, and it appears only in the second moment. That asymmetry is itself a finding: public attention co-moves with risk, not with direction.

Read together, the chapters carry a practical asymmetry for anyone trading silver: the level and direction of next week’s return are not forecastable from public information, but its risk is. That is exactly the input a mean-variance or value-at-risk framework consumes—position sizing, option pricing, and tail-risk budgeting all run on the conditional variance, not the conditional mean. The mean-variance framing of the title is in that sense not decorative: the forecastable moment is the one risk management actually uses.

9 Conclusion

Weekly silver returns are indistinguishable from a martingale difference under magnitude loss, consistent with weak- and semi-strong-form efficiency in the economically relevant sense. The residual directional traces are weak, model-dependent and unexploitable; their apparent temporal clustering is consistent with—though not proof of—the adaptive-markets view (Lo, 2004); the Markov volatility states do not explain when the directional windows arise. The conditional variance, in contrast, is strongly forecastable: HAR, GARCH and the tree ensembles all beat the naïve realised-volatility floor under the proxy-robust QLIKE loss, attention and sentiment add a small but statistically solid increment, and the VaR backtest shows the gain is economically material. Together the

two verdicts form the thesis’s single statement: efficiency binds the conditional mean and leaves the conditional variance free—and the forecastable moment is the one risk management actually consumes.

Contributions. The thesis makes four contributions. First, it evaluates a full model ladder—classical, machine-learning, and deep-learning—for *both* moments of weekly silver returns under one drift benchmark and one significance battery, so that model families are compared on a level footing. Second, it makes the conditional-mean / conditional-variance factorisation explicit, turning what could read as “returns failed, volatility worked” into a single coherent efficiency statement. Third, it builds a domain-appropriate sentiment pipeline—Twitter-RoBERTa for Reddit, FinBERT for financial news—and treats sentiment as a falsifiable ablation on both moments rather than assuming it helps. Fourth, it documents a carefully qualified set of negative results on semi-strong predictability of the mean, separating magnitude from direction and showing where an apparent directional signal fails to replicate across model classes.

Limitations. Several caveats bound these conclusions. First, the evidence is one asset over one decade (591 weeks, 2015–2026) with a 174-week out-of-sample window that the exceptional 2025 bull dominates; a failure to reject efficiency is not proof of it, and the power to detect a small predictable component in a heavy-tailed weekly return is limited. Second, sentiment and attention remain model-scored proxies drawn from a bounded set of channels—reputable English news (paid, full-text), multilingual GDELT headlines, Reddit, and Google-Trends search—so an effect carried by sources or platforms outside this set, or one a sharper measure of tone would catch, could be missed. Third, the entire study is at the weekly horizon: efficiency could bind less tightly, or predictability surface, intraday or monthly, and the daily-input siblings only partly probe this while still targeting the weekly return. Fourth, realised volatility is built from daily squared returns rather than intraday data—a coarse proxy for the true integrated variance.

Future work. Each limitation points to an extension. Higher-frequency (five-minute) realised variance would sharpen the volatility target and let the mean be re-tested intraday, where microstructure may admit fleeting predictability the weekly view averages away. Because the Markov volatility states do not explain the rolling directional episodes (Section 6), a direct time-varying-coefficient model or a model whose states are defined by investor attention rather than return variance would be a more informative test of the adaptive-markets interpretation. The VaR backtest of Section 7.4 is also a first pass; a cost-aware volatility-managed strategy backtested net of transaction costs would turn the tail-risk result into a fuller economic-value study. Finally, applying the same two-moment framework to other metals—gold, platinum, copper—would show whether weekly efficiency is specific to silver or shared across liquid metal markets.

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A Data appendix

Table 7 catalogues every variable entering the analysis—its source, native frequency, and how it enters the models. The notes below detail each source family: the choices a summary table cannot convey.

Table 7: Variables entering the analysis, by feature group. All features are observed at the Friday- $t-1$ close and lagged one week by each model before use, except the macro block (publication-lagged) and a best-lag ablation set. “EXOG” denotes the six cross-asset returns.

Group	Variables	Source	Freq.	Entered as
Target	silver_return	Yahoo Finance	Daily	weekly W-FRI log-return
AR	silver_lag1/2/3	Yahoo Finance	Daily	weekly log-returns, 1–3 wk lags
Cross-asset (EXOG)	gold_return, copper_return, usd_return, sp500_return, vix_return, oil_return	Yahoo Finance	Daily	weekly log-returns
Ratio (GS)	gs_ratio_z	Yahoo Finance	Daily	gold/silver ratio, z-scored on train
FRED daily	real_rates_chg, breakeven_chg, jobless_chg	FRED	Daily/Wk	weekly change
Positioning (COT)	cot_mm_net_pct, cot_comm_net_pct	CFTC	Weekly	net position / open interest
Sentiment	reddit_sentiment, news_sentiment	Reddit, GDELT	Daily	weekly mean of RoBERTa / FinBERT scores
Paid news (NEWS_PAID)	news_paid_sentiment_title	NewsAPI.ai	Daily	weekly mean of FinBERT title scores
Technical (TECH)	macd_line, macd_hist, rsi_14w, mom_5w, roc_13/26/52w, price_ma13w, donchian_52w, bb_pct_b	Yahoo Finance	Weekly	directional momentum / trend indicators
Macro (MACRO)	cpi, fed_funds, ind_prod, m2	FRED	Monthly	3 publication-availability lags each
PMI	china_pmi_proxy	FRED	Monthly	latest published monthly level

A.1 Sources

- **Market prices** — Yahoo Finance, daily. The three metals are global benchmarks; the equity, dollar, volatility, and oil series are US-centric:
 - silver (**SI=F**): COMEX front-month futures, the global silver benchmark — the forecast target
 - gold (**GC=F**): silver’s closest monetary substitute and dominant co-mover
 - copper (**HG=F**): industrial-demand proxy (about half of silver demand is industrial)
 - US dollar index (**DX-Y.NYB**): the dollar against a basket of major currencies; metals are dollar-priced, so a stronger dollar weighs on silver
 - S&P 500 (**^GSPC**): US large-cap equities — broad risk appetite
 - VIX (**^VIX**): 30-day implied volatility from S&P 500 options — the market’s fear / risk-off gauge
 - oil (**CL=F**): WTI crude, the US benchmark — commodity-complex and inflation proxy

The six cross-asset weekly log-returns form the EXOG set (Batten et al., 2010; Zhang & Wei, 2010).

- **Rates and inflation** — FRED, US, daily:
 - 10-year TIPS real yield (**DFII10**): the inflation-adjusted return on 10-year US Treasury bonds—the “real” interest rate. Silver pays no income, so a higher real yield makes interest-bearing bonds the more attractive store of value and tends to pull the metal down.
 - 10-year breakeven inflation (**T10YIE**): the average inflation rate markets expect over the next decade (the gap between ordinary and inflation-protected 10-year Treasury yields). Silver is held as an inflation hedge, so rising expectations can support it.
 - initial jobless claims (**ICSA**): the number of new US unemployment-benefit filings each week—a timely read on the labour market and the economy’s direction, with rising claims signalling a weakening economy.

Each enters as a weekly change.

- **Macroeconomic** — FRED, US, monthly:
 - Consumer Price Index (**CPIAUCSL**): the headline measure of realised US consumer-price inflation
 - federal funds rate (**FEDFUNDS**): the Fed’s overnight policy interest rate, which drives real yields and the dollar

- industrial production (INDPRO): the Fed’s index of US factory, mining, and utility output; about half of silver demand is industrial, so stronger output lifts physical demand for the metal
- M2 money supply (M2SL): a broad measure of money in circulation (cash plus bank deposits and near-money); rapid growth signals future inflation, which tends to lift hard assets like silver

These enter under a point-in-time availability rule rather than a naive shift: a value is visible only once it would actually have been released, with three publication lags kept per series.

- **Positioning** — CFTC (US) Commitments of Traders, COMEX silver, weekly:
 - managed-money net positioning: speculative (fund) sentiment
 - commercial net positioning: hedger and producer positioning

Each as a share of open interest.

- **Business sentiment (PMI)** — FRED, monthly: a China business-confidence proxy (CHNBSCICP02STSAM), a balance-of-opinion level standing in for industrial demand from silver’s largest physical consumer. Unlike the macro block (three pre-lagged publication lags), it enters as a single point-in-time level—the most recent value released by its publication date (a 31-day reporting lag)—taken as the week’s last observation and lagged one week by each model, exactly like the gold/silver ratio. It is kept as its own ablation group rather than folded into the US macro set.
- **Sentiment** — Reddit, GDELT, NewsAPI.ai, Google Trends (English-language / worldwide):
 - Reddit posts (r/WallStreetSilver, r/Silverbugs; via Arctic Shift): retail attention and tone, scored with Twitter-RoBERTa (Loureiro et al., 2022)
 - GDELT news headlines: global media tone, scored with FinBERT (Araci, 2019)
 - NewsAPI.ai (Event Registry) paid news: reputable English-language sources, from 2015 and at least one article per week, scored with FinBERT (Araci, 2019) on the article title
 - Google Trends “silver” search interest: worldwide retail attention (volatility chapter only)

Each text source is averaged to a weekly mean. GDELT is sparse—it begins only in late 2017 and has zero-article weeks even inside the test window—so the news series is a documented limitation; the paid NewsAPI.ai feed covers the full

sample and is the cleaner news source, while the volatility chapter (Section 7) draws on Reddit, paid news, and Google Trends instead. From each text feed that chapter derives the weekly attention and tone-intensity features of Section 4.2, plus a within-week tone-dispersion measure (the standard deviation of daily tone) for Reddit and paid news; dispersion is dropped from every ablation rung after exploratory analysis found no correlation with realised volatility.

A.2 Technical-indicator formulas

Every indicator in the TECH group (Section 4) is a transform of the weekly (W-FRI) silver close P_t , with $r_t = \log(P_t/P_{t-1})$ the weekly log-return. Write EMA_k and SMA_k for the exponential and simple moving averages of span/window k weeks, σ_k for the rolling k -week standard deviation, and $\min_k, \max_k$ for the rolling k -week low and high; RSI uses exponentially-smoothed average gains and losses. Each is built raw and lagged one week before use.

Table 8: Technical-indicator definitions (weekly silver close P_t).

Indicator	Definition	Window (wk)
MACD line	$\text{EMA}_{12}(P_t) - \text{EMA}_{26}(P_t)$	12, 26
MACD histogram	$\text{MACD}_t - \text{EMA}_9(\text{MACD}_t)$	9
RSI	$100 - \frac{100}{1 + \text{RS}_t}, \quad \text{RS}_t = \frac{\text{EMA}_{14}(\text{gains})}{\text{EMA}_{14}(\text{losses})}$	14
Momentum	$\sum_{i=0}^4 r_{t-i}$	5
Rate of change	$P_t/P_{t-k} - 1, \quad k \in \{13, 26, 52\}$	13, 26, 52
Price-to-MA	$P_t/\text{SMA}_{13}(P_t) - 1$	13
Donchian position	$\frac{2(P_t - \min_{52})}{\max_{52} - \min_{52}} - 1$	52
Bollinger %B	$\frac{P_t - (\text{SMA}_{20} - 2\sigma_{20})}{4\sigma_{20}}$	20

B Supplementary figures

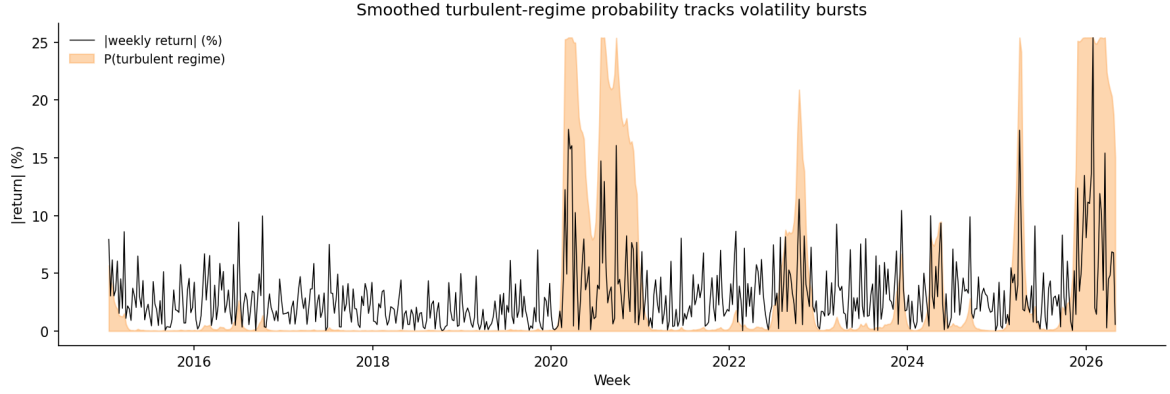


Figure 8: Validation of the Markov volatility-state interpretation. The black line is the absolute weekly silver return; the orange area scales the smoothed probability of the high-variance turbulent state to the same vertical range. The probability rises around the largest return magnitudes, supporting the calm/turbulent labels independently of the rolling directional test. Referenced from Section 6.

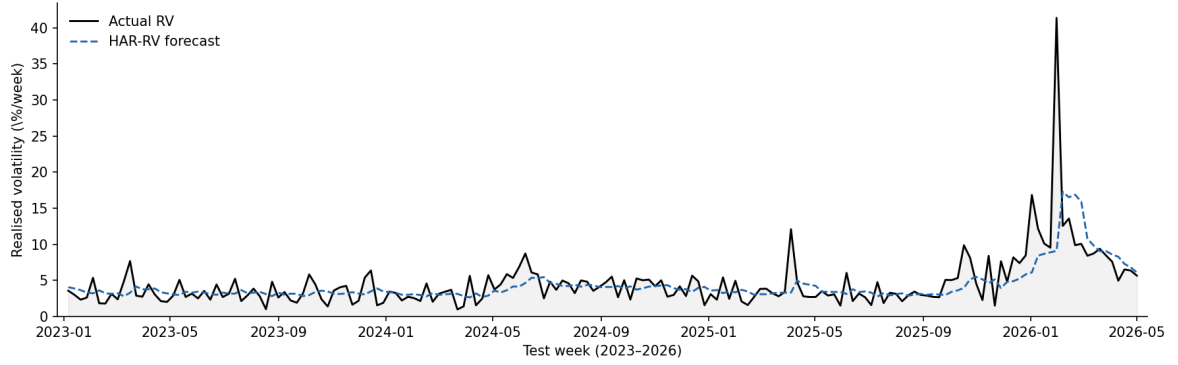


Figure 9: Weekly realised volatility over the full test window: the actual series (black) and the HAR-RV forecast (blue dashed). Unlike the return, the volatility is visibly persistent—it clusters—and the parsimonious HAR tracks both the calm 2023–2024 band and the 2025–2026 run-up, lagging only the sharpest single-week spike of early 2026. It is the visual mirror of the return chapter, where no model could pull away from its floor (Figure 3). Referenced from Section 7.